

6(a)	<u>human immunodeficiency virus</u> ;	1
6(b)	<i>max 1 if no mention of benefit of ART</i> <i>any two from:</i> helps to, increase / maintain / AW, number of, T-,lymphocytes / T-helper cells ; A helps to prevent death of T-lymphocytes / AW A T-cells maintains / improves / AW, strength of, immune system / immune response ; increases chance of recovery from, infectious / opportunistic, disease ; AW <i>idea that</i> decreases risk of having / helps prevent, other (opportunistic) diseases ; AVP ; e.g. (can) decrease viral load / reduce number of HIV	2
6(c)(i)	X is positioned <u>above</u> the join on the coronary artery on, left-hand side of Fig. 6.1 / right heart ; R if placed at the junction of the vein and artery	1

Question	Answer	Marks
6(c)(ii)	can withstand high(er) blood pressure / less likely to burst ; R collapse A <i>idea that</i> blood pressure is now higher (so) blood vessel may last longer ; A otherwise unlikely to last a long time (in tunica media) more (smooth) muscle / thicker (smooth) muscle layer / more elastic fibres ; detail ; e.g. helps, onward movement / faster flow, of blood / AW A otherwise does not have enough, (smooth) muscle / elastic fibres, for onward movement / required speed of flow, of blood thicker tunica, externa / adventitia ; A more collagen	2